How-To Library

Informatica

Data Engineering Integration with a WANdisco-enabled Cluster (10.4.1)

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Abstract

WANdisco fusion is a software application that replicates HDFS data among cluster nodes that are running different versions or distributions of Hadoop to prevent data loss in case a cluster node fails.

You can use Informatica Data Engineering on Cloudera or Hortonworks clusters where WANdisco is enabled. This article describes how to configure the environment and Data Engineering Integration on WANdisco-enabled clusters.

Supported Versions

- Data Engineering Integration 10.4.1

Table of Contents

Overview. 2
Prerequisites. 2
Enable the Domain in WANdisco-enabled Cluster (Hortonworks). 2
Enable the Domain in WANdisco-enabled Cluster (Cloudera). 3

Overview

You can use Informatica Data Engineering Integration with a WANdisco-enabled cluster. When the cluster runs WANdisco Fusion, HDFS data in cluster storage resources is replicated across cluster nodes.

Prerequisites

Verify the following prerequisites.

Informatica Prerequisites

Install Informatica Data Engineering Integration 10.4.1 on-premises or in a cloud environment. For example, Data Engineering Integration could be co-located in a cloud environment with a supported Hadoop cluster.

Cluster Prerequisites

Integrate WANdisco Fusion version 2.14.2.1 on a cluster with one of the following Hadoop distributions:

- Cloudera CDH 6.3 on RedHat 7.4 and SUSE 12.3
- Hortonworks HDP 3.1.4 on RedHat 7.4 and SUSE 12.3

Enable the Domain in WANdisco-enabled Cluster (Hortonworks)

Copy WANdisco Fusion .jar files to all machines that host the Data Integration Service:

1. On the Hortonworks cluster headnode, copy the following files:
   a. Browse to the following directory: `/opt/wandisco/fusion/client/jars` and copy all of the .jar files.
   b. Browse to the following directory:`/usr/hdp/<version>/hadoop` and copy the following files:
      - `hadoop-common-<version number>-hdp<version number>.jar`

- hadoop-auth-<version number>-hdp<version number>.jar

c. Browse to the following directory: /usr/hdp/<version>/hadoop-yarn-client and copy the following files:

- hadoop-yarn-api-<version number>-hdp<version number>.jar
- hadoop-yarn-common-<version number>-hdp<version number>.jar
- hadoop-yarn-client-<version number>-hdp<version number>.jar

2. Back up and delete the following files from each machine that hosts the Data Integration Service:

a. Back up and delete the following files from the <Informatica home>/services/shared/hadoop/<distribution version>/lib directory:

- hadoop-yarn-api-<version number>.jar_bck
- hadoop-yarn-common-<version number>.jar_bck
- hadoop-yarn-client-<version number>.jar_bck
- hadoop-common-<version number>.jar_bck
- hadoop-auth-<version number>.jar_bck

b. Back up and delete the following files from the <Informatica home>/services/shared/spark/<Spark version> directory:

- hadoop-yarn-api-<version number>.jar_bck
- hadoop-yarn-common-<version number>.jar_bck
- hadoop-yarn-client-<version number>.jar_bck
- hadoop-common-<version number>.jar_bck
- hadoop-auth-<version number>.jar_bck

3. Paste the .jar files that you copied to the following locations on each machine that hosts the Data Integration Service:

- To run jobs using the Blaze engine, paste the .jar files to <Informatica home>/services/shared/hadoop/<distribution version>/lib.
- To run jobs using the Spark engine, paste the .jar files to <Informatica home>/services/shared/spark/<Spark version>/.
- Add the following property in the Hadoop_Connection: spark.sql.hive.metastore.jars=./infa_rpm.tar/services/shared/hadoop/HDP_3.1/infaLib/*:./infa_rpm.tar/services/shared/hadoop/HDP_3.1/spark/standalone-metastore/*::/opt/wandisco/fusion/client/lib/*

4. Recycle the Data Integration Service on each node.

Enable the Domain in WANdisco-enabled Cluster (Cloudera)

Copy WANdisco Fusion .jar files to all machines that host the Data Integration Service:

1. On the Cloudera cluster headnode, copy the following files:

a. Browse to the following directory: /opt/cloudera/parcels/FUSION-<version number>/lib/ and copy all of the .jar files.

b. Browse to the following directory:`/opt/cloudera/parcels/CDH-<version number>/jars` and copy the following files:

- `hadoop-yarn-api-<version number>-cdh<version number>.jar`
- `hadoop-yarn-common-<version number>-cdh<version number>.jar`
- `hadoop-yarn-client-<version number>-cdh<version number>.jar`
- `hadoop-common-<version number>-cdh<version number>.jar`
- `hadoop-auth-<version number>-cdh<version number>.jar`

2. Back up and delete the following files from each machine that hosts the Data Integration Service:

a. Back up and delete the following files from the `<Informatica home>/services/shared/hadoop/<distribution version>/lib` directory:

- `hadoop-yarn-api-<version number>.jar_bck`
- `hadoop-yarn-common-<version number>.jar_bck`
- `hadoop-yarn-client-<version number>.jar_bck`
- `hadoop-common-<version number>.jar_bck`
- `hadoop-auth-<version number>.jar_bck`

b. Back up and delete the following files from the `<Informatica home>/services/shared/spark/<Spark version>` directory:

- `hadoop-yarn-api-<version number>.jar_bck`
- `hadoop-yarn-common-<version number>.jar_bck`
- `hadoop-yarn-client-<version number>.jar_bck`
- `hadoop-common-<version number>.jar_bck`
- `hadoop-auth-<version number>.jar_bck`

3. Paste the .jar files that you copied to the following locations on each machine that hosts the Data Integration Service:

- To run jobs using the Blaze engine, paste the .jar files to `<Informatica home>/services/shared/hadoop/CDH-<distribution version>/lib`
- To run jobs using the Spark engine, paste the .jar files to `<Informatica home>/services/shared/hadoop/CDH-<distribution version>/spark/jars/`

4. Recycle the Data Integration Service on each node.

Author

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